

PROJECT PROPOSAL

IT Asset & Inventory Log System

A Database Management System (DBMS) Project Proposal

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Department: Bachelor of Science in Computer Science and Engineering (B.Sc. in CSE)

Project Type: Database Management System (DBMS) Project

Duration: 6 – 8 Weeks

Version: 1.2

Submission Date: AUG 07, 2026

(Week 2 progress)

Submitted in partial fulfilment of the requirements of the Database Management Systems course

Database Design

The system is supported by a relational database consisting of three core tables, described below.

1 Table: Users

Field	Type	Description
UserID	INT (PK)	Unique identifier for each user
FullName	VARCHAR(100)	Full name of the employee
Department	VARCHAR(50)	Department the user belongs to
Email	VARCHAR(100)	Contact email address

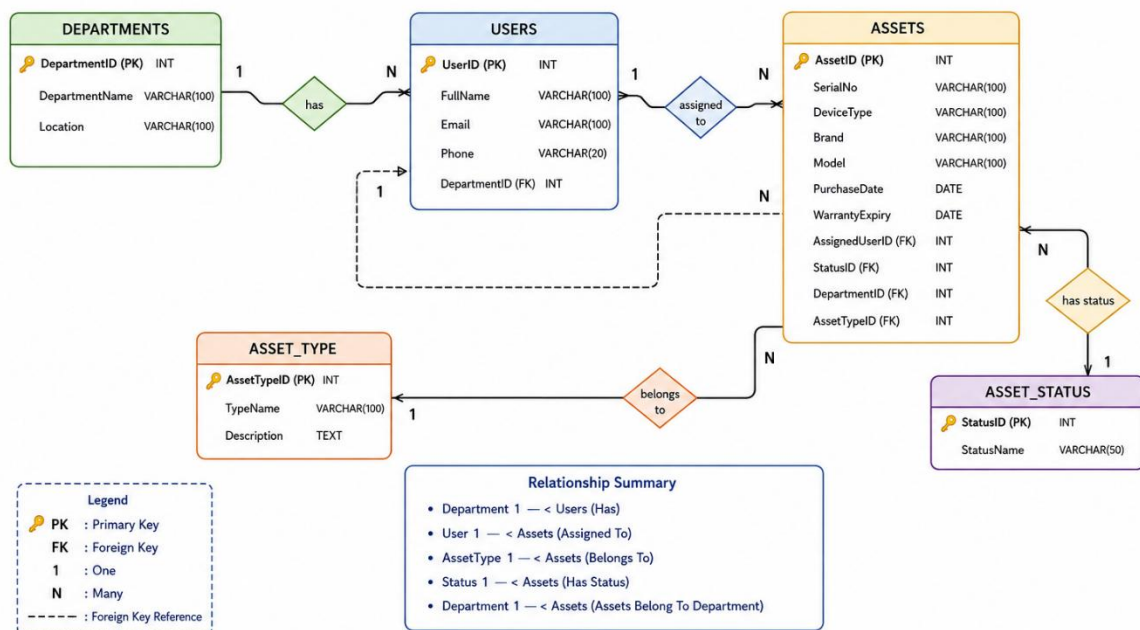
2 Table: Assets

Field	Type	Description
AssetID	INT (PK)	Unique identifier for each asset
SerialNo	VARCHAR(100)	Manufacturer serial number
DeviceType	VARCHAR(50)	Category of hardware
Brand	VARCHAR(50)	Manufacturer / brand name
Model	VARCHAR(50)	Model number or name
PurchaseDate	DATE	Date of purchase
AssignedUserID	INT (FK)	References Users.UserID
Status	VARCHAR(30)	Operational / Under Repair / Retired

Entity Relationship Diagram (ERD)

The diagram below illustrates the relationships between the Users, Assets, and Log tables. Each user may be assigned multiple assets and each asset may have multiple records .

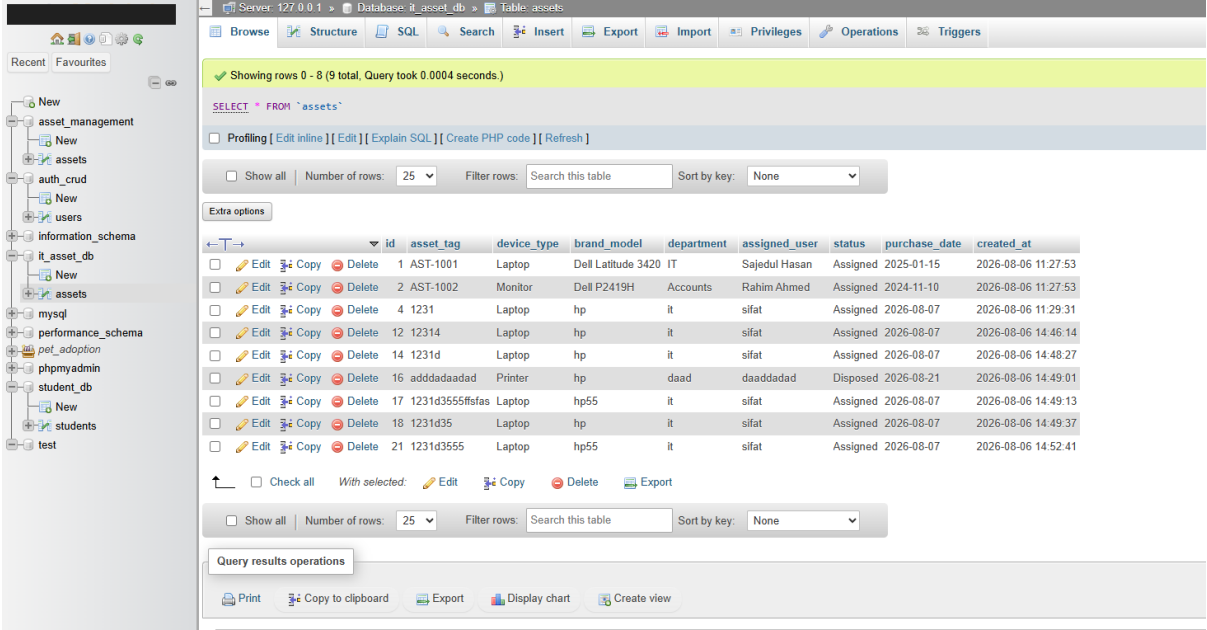
ER DIAGRAM OF IT ASSET INVENTORY LOG SYSTEM



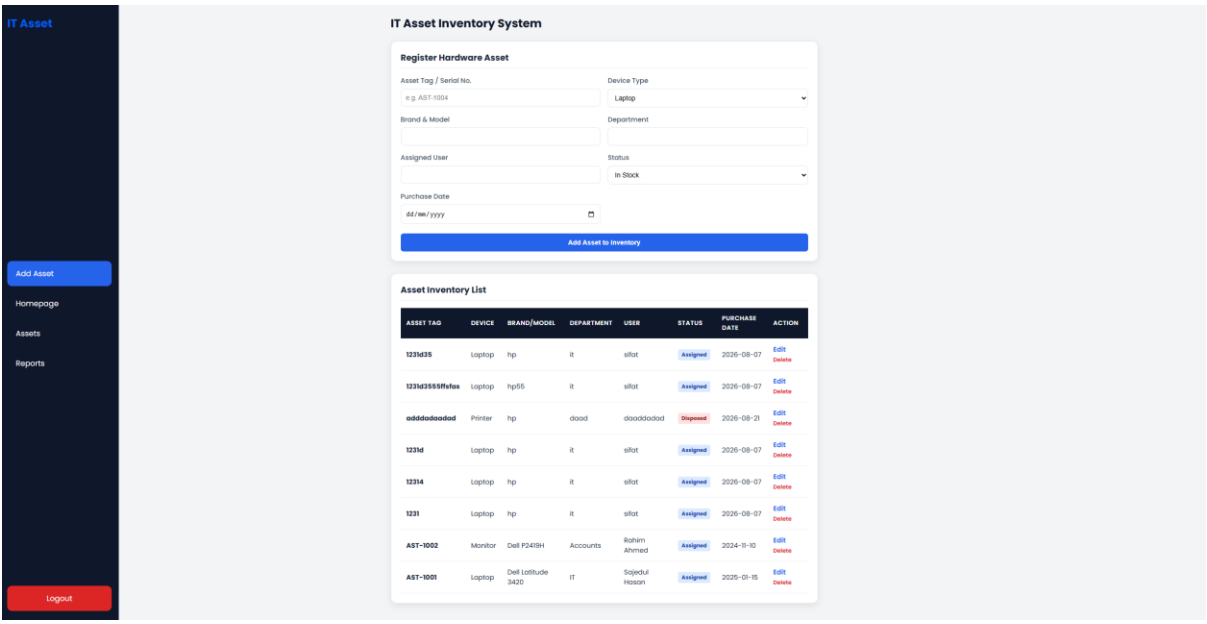
Technology Stack

Component	Technology
Frontend	HTML5
Styling	CSS3
Programming (Client-side)	JavaScript
Backend	PHP
Database	MySQL
Web Server	XAMPP
IDE	Visual Studio Code

Set up XAMPP & Create Database



Website Interface



PHP logic for Insert, View, and Update

```
File Edit Selection View Go Run ... sifai_bhaji_database
index.php index.php sidebar.css README.md style.css
index.php
1 <?php
2 $host = "localhost";
3 $user = "root";
4 $pass = "";
5 $dbname = "it_asset_db";
6
7 $conn = new mysqli($host, $user, $pass, $dbname);
8
9 if ($conn->connect_error) {
10 | die("Database Connection Failed: " . $conn->connect_error);
11 }
12
13 $message = "";
14
15 // Handle Delete Request
16 if (isset($_GET['delete'])) {
17 | $delete_id = intval($_GET['delete']);
18 | $stmt = $conn->prepare("DELETE FROM assets WHERE id = ?");
19 | $stmt->bind_param("i", $delete_id);
20 | if ($stmt->execute()) {
21 | | $message = "<p style='color: #166534; font-weight: bold;'>Asset removed successfully.</p>";
22 | }
23 | $stmt->close();
24 }
25
26 // Handle Form Submission (Insert)
27 if ($_SERVER["REQUEST_METHOD"] == "POST" && !isset($_POST['update_asset'])) {
28 | $asset_tag = trim($_POST['asset_tag']);
29 | $device_type = trim($_POST['device_type']);
30 | $brand_model = trim($_POST['brand_model']);
31 | $department = trim($_POST['department']);
32 | $assigned_user = !empty($_POST['assigned_user']) ? trim($_POST['assigned_user']) : 'Unassigned';
33 | $status = trim($_POST['status']);
34 | $purchase_date = $_POST['purchase_date'];
35
36 | $stmt = $conn->prepare("INSERT INTO assets (asset_tag, device_type, brand_model, department, assigned_user, status, purchase_date)
37 | $stmt->bind_param("sssssss", $asset_tag, $device_type, $brand_model, $department, $assigned_user, $status, $purchase_date);
38
39 | if ($stmt->execute()) {
40 | | $message = "<p style='color: #166534; font-weight: bold;'>Asset registered successfully!</p>";
41 | } else {
42 | | $message = "<p style='color: #991b1b; font-weight: bold;'>Error: " . $conn->error . "</p>";
43 | }
44 | $stmt->close();
45 }
46
47 // edit reqst
48 // =====
49 // EDIT & UPDATE
50 // =====
51
52 $editMode = false;
53 $editData = [
54 | 'id' => '',
55 | 'asset_tag' => '',
56 | 'device_type' => 'Laptop',
57 | 'brand_model' => '',
58 | 'department' => '',
59 | 'assigned_user' => '',
60 | 'status' => 'In Stock',
61 | 'purchase_date' => ''
62 ];
63
64 // Load data for editing
65 if (isset($_GET['edit'])) {
66 |
```